

(W5)

Q#96/ RidgeClassifier is a classifier variant of which regressor?
Ridge regressor.

Q#97/ The first step to apply Ridge classifier should be
Instantiate a classification estimator, without passing
any arguments.

Q#98/ What is a solver?

In the context of machine learning, a "solver" refers to an algorithm or method that is used to find the optimal solution to a mathematical problem.

Q#99/ What is the ^{solver} solver used for `scipy.linalg.solve` to min cost function?
cholesky.

Q#100/ Which solver can be used for RidgeClassifier for large data?
sparse-cg.

(*)

Perceptron: A Binary linear classifier, an algorithm that makes predictions based on a linear combination of input features.

Page No.:

Date:

(*)

SGD: Stochastic Gradient Descent.

↳ Optimization algorithm.

↳ Gradient Descent Algo: The goal is to iteratively move towards minimum of loss function.

The update rule involves adjusting model parameters in the opposite direction of the gradient of the loss function.

↳ Stochastic Gradient

Descent Algo: Instead of performing gradient descent on the entire dataset, SGD computes gradient on subsets of the data iteratively; makes each iteration of optimization ^{process} faster but introduces more noise to the updates.

Q#101/ *

Perceptron classification shares the same underlying implementation with SGD classifier. TRUE

Q#102/ *

Partial-fit method is a way to plot the iteration vs loss curve. TRUE

Q#103/ *

Single Label Multiclass classification has how many output labels? 1

Q#104/ *

OVR classifier supports ~~multiclass~~ multilabel classification. TRUE

Q#105/ *

Which stratified APIs are provided by sklearn.model_selection module to create folds?

✓ StratifiedKFold

✓ StratifiedShuffleSplit

Q#106) * Confusion matrix evaluates classification accuracy by computing the confusion matrix with each row corresponding to the true class.

TRUE

Page No: _____

Date: _____

Q#107) Confusion matrix can be displayed with which API in sklearn.metrics

ConfusionMatrixDisplay

Q#108) Which ^{Naive Bayes} NB is used to implement Gaussian Naive Bayes algorithm for classification? GaussianNB

* MNIST Modified National Institute of Standards and Technology

Q#109) The data matrix or the label vectors given as pandas dataframe can be converted to numpy array by which of the following method?
to_numpy()

Q#110) What is the way to ensure the used sklearn version at least 0.20?

```
import sklearn
assert sklearn.__version__ >= "0.20"
```

Q#111) What are true positives (TP)?
↓
matched predicted = True
↓
predicted label
↘
actual = predicted = +ve

Q#112) What is the formula for calculating recall?

$$\frac{TP}{TP + FN}$$

Q#113) What is the formula for F₁ score?

$$\frac{2}{(\text{precision})^{-1} + (\text{recall})^{-1}}$$

Q#114) What is the formula for Precision?

$$\frac{TP}{TP + FP}$$

Q#115) What is the formula of (accuracy)?

$$\frac{TP + TN}{\# \text{ Total-obs}}$$

Page No :
Date :

Q#116) What is the use of (precision recall curve)?

computes precision-recall pairs for different probability thresholds.

Q#117) * Combining multiple binary classifiers and devising a suitable scoring metric are used to extend the idea of perceptron to handle multiclass problems?

TRUE.

Q#118) Perceptron is a part of which library module?
sklearn.linear_model.

Q#119) The class of a perceptron that outputs the max score for the input sample is considered as predicted class

Q#120) Can GridsearchCV be used to improve the learning rate of the Perceptron class?
Yes, certainly.

Q#

Q#121) Is the following true about Naïve-Bayes classifier...

$$P(x_i | y, x_1, x_2, \dots, x_{i-1}, x_{i+1}, \dots, x_m) = P(x_i | y)$$

TRUE

Q#122) Which of the following modules of sklearn contains the API GaussianNB?

sklearn.naive-bayes

Q#123) Consider the following code snippet.

```
from sklearn.naive-bayes import GaussianNB
```

```
gnb = GaussianNB()
```

```
gnb.fit(x, y)      when does gnb best suit for X?
```

gnb is best suited for X, if it does not have any categorical features

Page No:

Date:

Q#124) Which of the following is best suited for imbalanced data?

* Q#124) Which of the following is best suited for imbalanced data?

ComplementNB

Q#125) Which of the following Naive Bayes classifiers is suitable for classification of word counts for text classifications?

MultinomialNB

Q#126) How many classes are there in MNIST dataset?

(10)

Q#127) Let X be a feature matrix with shape (1000, 5) and y be the label vector with two classes: 0 and 1. Assume that 650 examples of training data belong to class 0.

What will be the output of the following code?

```
base_clf = DummyClassifier(strategy='most-frequent')
base_clf.fit(X, y)
print(base_clf.score(X, y))
```

650 training (class 0)

$$\text{base_clf_score} = \frac{650}{1000} = 0.65$$

Q#128) Which of the following are true about DummyClassifier?

✓ DummyClassifier makes predictions that ignore the input features

✓ DummyClassifier serves as a simple baseline to compare against more complex classifiers

✓ The predictions of DummyClassifier typically depend on values observed in the y parameter passed to

used as a simple baseline or reference model; not intended for real world scenarios but serves as a useful tool for understanding the performance of more sophisticated classifiers

basically a 50/50

Dummy baseline performance $\text{fit}()$ metric based on the distribution of classes of training dataset, $\text{predict}()$ for comparing better performance of model

Q#129)

Consider the following classifier and ^{answer} select the correct option ~~only~~ application it pursues.

Page No :
Date :

estimator = SGDClassifier(loss='log', penalty='l2',
max_iter=1, warm_start=True, class_weight=None,
alpha=0, learning_rate='constant',
random_state=1729)

Logistic loss
associated with
logistic regression

Q#130)

Which sklearn module contains API for Logistic Regression?
sklearn.linear_model.

Q#131)

What is the loss function used in SGDClassifier for Logistic regression?
log.

*

Solver: Solver refers to the algorithm or optimization method that is opted to minimize the cost or loss function during the training of a model.

Q#132)

Which solver in LogisticRegression() is a better choice for a small dataset?
liblinear

Q#133)

Which API is used to automatically set the best hyperparameters in logistic regression?
LogisticRegressionCV()

Q#134)

Which parameter in LogisticRegressionCV() stores the list of ~~parameters~~ values for regularization parameters?
C

What is its "default value"? 10

Q#135)

In Ridge Classifier, what is the default penalty?
L2 penalty.

Q#136) Which parameter in ~~sklearn.cross~~ ~~sklearn.model.selection~~ ~~cross-validate~~ is set to True to get the training scores?
`return_train_score`

Page No :
 Date :

Q#137) Which of the following method is used to estimate probabilities in SGD Classifier with suitable loss?

`predict_proba`

Q#138) Which method of classification needs 'n' classifiers, where n is the number of classes?

`OneVsRestClassifier`

Q#139) If we classify the MNIST dataset with `OneVsRestClassifier` how many classifiers will be learnt? 10

Q#140) What is the API that converts text input into a vector of numerical values?

`TfidfVectorizer`

Q#141) How many classes are there in 20 newsgroup data?
 20

Q#142) Which attribute is used to get the no of samples encountered for each class during naive bayes classifier fitting?

`class_count`

Q#143)

cg	344	13	32	0
ss	6	361	24	0
src	1	5	392	0
trm	4	12	187	0
	cg	ss	src	trm

Mention no of misclassifications of ss & src & # of misclf of src as of ss based on the adjoining confusion matrix?

$$\begin{aligned} \text{ss-misclf} &= 6 + 24 = 30 \\ \text{src-misclf} &= 1 + 5 = 6 \\ \text{src-ss-misclf} &= 5 \end{aligned}$$

* `asFrame = True` → parameter used while loading datasets from sklearn in order to retrieve data as a dataframe.

W7

- Softmax Regression algorithm is used for solving multi-class classification.
- Softmax function normalizes as input vector into a probability distribution.

Page No :

Date :

Q#144) To solve multiclass classification problem using logistic-Regression estimator we set multi-class parameter to multinomial

~~Q#145) Which are the different algorithms?~~

Q#145) Given confusion matrix

Find accuracy score

Actual	F	21	842
	T	939	11
		T	F
		Predicted	

Accuracy score

$$= \frac{TP + TN}{\# \text{ Total}}$$

$$\text{Accuracy} = \frac{939 + 842}{939 + 842 + 21 + 11} = \frac{1781}{1813} = 0.98$$

Q#146) Where does confusion-matrix method lie in sklearn modules?
sklearn.metrics

Q#147) Which sklearn objects are useful in discovering best value of K for a KNN model?

✓ GridSearchCV

✓ RandomizedSearchCV

✓ validation-curve.

Q#148) Which of the following sklearn module implement relevant functions for the Nearest Neighbors algorithm?

sklearn.neighbors.

(*)

let N denote number of training data points. For every dataset, $\exists$ a threshold N_0 such that the increasing value of N beyond this threshold does significantly affect the accuracy of the model.

Q#149) What does the value k in k -Nearest Neighbors algorithm refer to? Number of k -nearest neighbors No. to consider for each test data point. Date: _____

- ★ k nn models with low values of K produce complex boundaries
- ★ k nn models with high values of K produce simple boundaries

Q#150) If $x_1, x_2, \dots, x_k$ are k nearest data points of x , then $\hat{y}(x) = ?$

$$\hat{y}(x) = \frac{1}{k} \sum_{i=1}^k y(x_i)$$

- The number k in KNN is a parameter to be estimated from the training data.
- The time ~~to~~ complexity depends upon #training-instances & not on dimension

Q#151) Which of the following sklearn objects are relevant to perform classification using k -Nearest Neighbors?

KNeighborsClassifier

Q#152) What is true about ~~the~~ weights parameter of KNeighborsClassifier() class?

It represents the relative weights assigned to all neighboring point as a function of their distance from the test data point, which are to be considered during majority vote.

Q#153) Performance of brute force KNN algo depends on?

- ✓ Chosen distance metric
- ✓ no of training pts

Q#154) Which sklearn object helps us find the k -nearest neighbors of a test point within a given training set?

- ✓ NearestNeighbors
- ✓ KNeighborsTransformers

Q#155) Match them ^{based on #training, $df > 100$}

I. $k=1$ a. ~~Overfitting~~ Overfitting
b. Underfitting
c. low var
d. High var

II. $k=2$ a. ~~Overfitting~~ Overfitting
b. Underfitting
c. low var
d. High var

(I) $k=1 \rightarrow$ overfitting
 $\rightarrow$ High variance

(II) $k=100 \rightarrow$ underfitting
 $\rightarrow$ low variance

Page No: _____
Date: _____

Q#156) What can be used to evaluate performance of KNN model?

- ✓ Accuracy score
- ✓ Misclassification error
- ✓ Root Mean Squared Error
- ✓ R^2 score

- ① It is essential to perform normalization before performing KNN classification or regression.
- ② The distance metric used in KNN may be sensitive to the relative scales of the features, which may prove problematic for datasets having substantial discrepancies.
- ③ As k increases, the bias of KNN regression model increases, & variances decreases.
- ④ KNN models are highly sensitive to noise & outliers for low values of k .

Q#157) What methods can be used to handle large data files in machine learning?

- Stream data or use incremental learning
- Use Big Data platform like Hadoop & Spark

Q#158)

- ① Sklearn processes large data in batches and updates the model parameters for each batch. The aim of incremental learning is for the learning model to adapt new data without forgetting its knowledge.
- ② Incremental learning is a dynamic technique which is applied when training data becomes available.

gradually over time or its size is out of system memory limits.

Page No.:

Date:

Q#158

Which estimator implements partial-fit method in Regression: SGD Regressor

Q#159

Which estimator ~~of~~ does not implement partial-fit in classification: Logistic Regression.

* 1 Epoch: when one entire pass is done through ~~the~~ the entire dataset

* 1 iteration: one updation in the dataset.

* ~~one~~ Regularization parameter strength $\uparrow \Rightarrow$ $\downarrow$ C-value and lower chances of overfitting.

* SVM $\rightarrow$ we try to maximize margin.
• Linear SVM $\rightarrow$ $C \uparrow \rightarrow \downarrow$ margin $\rightarrow$ Overfitting $\rightarrow$ lesser training error
• SVC $\rightarrow$ support vector classifier
• NuSVC

* Test error $\propto \frac{1}{\text{Training error}}$

* Bernoulli NB: All ~~parameters~~ ~~as~~ features are to be in binary. (when feature follows Bernoulli)

* var-smoothing: portion of largest variance of all features that is added to variances for calc stability

* Gaussian NB: apply when ~~feature~~ feature follows Gaussian distribution.

* Overfitting: High variance $\therefore$ Underfitting: Low variance

$$\text{Var} \propto \frac{1}{\text{Bias}}$$

* Decision Tree tends to overfit as it follows Greedy Search Algo

* Complex Boundaries: ~~Overfitting~~ ^{fitting} $\therefore$ Smoother Boundary: Underfitting

* KNN $k \downarrow$: Overfitting $\therefore k \uparrow$: Underfitting
 $\rightarrow$ very high $\rightarrow$ length of data

Q#160

y_true = ["cat", "ant", "cat", "cat", "ant", "bird"]
y_pred = ["ant", "ant", "cat", "cat", "ant", "cat"]

Page No :

What will be the precision value for ^{prediction} _{ant} class?

Precision value for ^{prediction} _{ant} class = $\frac{\text{Total matches}}{\text{Total length}}$

	ant	bird	cat
0	0	1	2
1	2	0	0
2	0	0	1
3	1	0	2

Precision

$$= \frac{2}{3} = 0.67$$

Q#161) Which of the following code will throw an error?

- from sklearn import svm No error
- from sklearn import SVC Error
- from sklearn.svm import SVC No error
- from sklearn import svm.SVC.

Q#162) Which keyboard is used to set the Regularization parameter in svm.SVC? C

Q#163) Which of sklearn module contains API confusion matrix
sklearn.metrics

Q#164) What does svc_scores() return? mean accuracy on the given test data & labels

* SVM classifiers DO NOT output probabilities for output class

Q#165) Which of the following method is used to get the indices of support vectors? .support_

Q#166) Which method returns no of support vectors? .n_support_

Q#167) What is the default kernel used in NuSVC()? rbf
Radial Basis Function

Q#168) Which keyword is used in NuSVC to set an upper bound on the fraction of margin errors & lower bound of fraction of support vec? nu

Q#169) What is the use of 'gamma' parameter?

gamma: Kernel Coefficient

Page No:

Date:

Q#170) What will the following code implement?

```
pipe1 = Pipeline([('scaler', MinMaxScaler()), ('classifier', SVC(kernel  
= 'linear', C=1))])
```

```
pipe1.fit(X_train, y_train, ravel(), cv=2)
```

```
cross_val_score(pipe1, X_train, y_train, ravel(), cv=2)
```

It will give the accuracy of training data X_{train} which is learn using linear SVM

Q#171)

Decision Trees are ~~non~~ parametric unsupervised learning method used for classification & regression. TRUE

Q#172) What is maximum possible value for gini impurity? 0.5

* Decision Trees ~~have~~ follow greedy approach for model building.
↳ adds noise to data can result in a drastically different decision tree.

* Gini impurity, entropy are measures of impurity or disorder of a set of data points

* Decision Trees are referred to as "white box" methods because they are transparent and easy to interpret.
"black box" are complex and hard to interpret.

• We may choose to ignore scaling as Decision Trees do not require feature scaling.

• If there are less no. of samples, Decision Trees will overfit.

Q#173) By limiting the height of the decision tree, what happens to variance? variance decreases

Q#174) What can be the minimum no of samples in all leaves of DT once algo has terminated? 1

Q#175) Which are the plots that help us observe correlation among different features of a dataset?

sns. heatmap()
sns. pairplot
df. corr()

Page No.:

Date:

Q#176) Which function do we use for exporting feature names and threshold values used by sklearn.tree.DecisionTreeClassifier?

export_text(clf, features_names = list(feature_names))

Q#177) ? Default values

(a) min-samples-split: 2

(b) min-samples-leaf: 1

- Bagging can make unstable classifiers like Decision Trees more robust. TRUE
- ▶ When voting is set to soft, module uses pred class labels based on the argmax of the sums of the pred probabilities
- ▶ When voting is set to hard, module uses pred class labels by majority rule voting
- Random Forest differ from bagging by forcing the tree to use a subset of its available features while constructing each tree. TRUE

Q#178) What is the default value of n-estimators in RandomForestClassifier?

n-estimators: 100

Q#179) ? Default base estimator of rf? DecisionTreeClassifier

Q#180) ? parameter in RandomizedSearchCV is used to specify #parameter settings that are sampled?

n_iter

Q#181) In sklearn.ensemble.AdaBoostRegressor, subsequent regressors focus more on difficult cases. TRUE

Page No :

Date :

Q#182) What does keeping the value of subsample lesser than 1.0 do to sklearn.ensemble.GradientBoostingClassifier

results in Stochastic Gradient Boosting

~~Q#183)~~ • Absolute max ^{robust} >>> squared error for outliers

• Huber loss combines both abs mae & sq. error for a more robust loss funⁿ less impacted by outliers

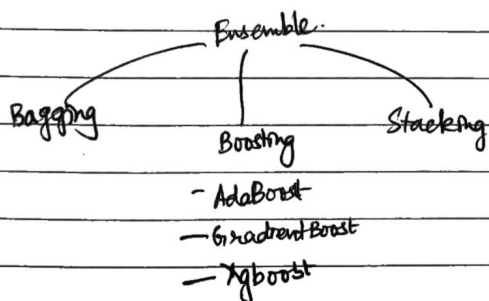
* Boosting $\rightarrow$ learner are trained sequentially

* Bagging $\rightarrow$ models are trained in parallel

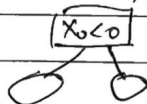
* $\exists$ a trade off b/w learning rate & n-estimators parameters

Q#183) In GradientBoostingRegressor, which parameter shrinks the contribution of each tree? learning rate

Q#184) In GradientBoostingRegressor, which parameter is used to specify the fraction of samples to be used for fitting the individual base learners? subsample.



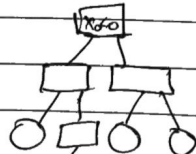
AdaBoost
IDC multiple stumps



Stump 1
 $\downarrow$
Stump 2 Higher weightage

uses feature

Gradient Boost



learning rate

• FeatureHasher present in sklearn.feature-extraction, which is a high speed, low-memory vectorizer that uses a technique known as feature hashing;

Q#185) What are some issues with KNN?

- computes distance of every point
- Prediction is computationally expensive
- No model is learnt
- We have to remember all the points for predicting.

Q#186) Which feature we should select for stumps?
classification {
 → Gini Index (criterion)
 → Entropy
 → squared error

Page No.:

Date:

Q#187) Convert pandas Series to numerical values

`b = pd.to_numeric(b, errors='coerce')`

Q#188) What is `errors='coerce'`?

It is error handling, it handles the errors during conversion from one datatype to another and if there rises some errors, it imputes the conversion as NaN.

Q#189) What are outliers?

Number of datapoints falling in the following range:

< 0.25 quantile or > 0.75 quantile

If `#outliers > 0`, I any outlier question.

Q#190) Count missing values
in dataframe

`df.isnull().sum()`